

Project Proposal

SentinelLite: A Lightweight SIEM with Rule-Based Detection and ML Anomaly Scoring

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Problem Statement

Small labs and student environments produce logs from web servers, Linux authentication services, firewall events, and Windows endpoints. These logs often contain evidence of scanning, brute-force login attempts, injection probes, and suspicious administrative activity. Heavy SIEM platforms can be costly or complex for this context. SentinelLite provides a lightweight SIEM that converts these logs into actionable alerts and anomaly findings.

Objectives

Objective	Expected Output
Multi-source ingestion	Apache, syslog/auth, firewall syslog, and Windows CSV parsing.
Rule detection	Failed logins, SQL injection, path traversal, scanner user agents, 404 probing, port
ML anomaly scoring	Isolation Forest scoring on 15-minute source-IP windows.
Analyst workflow	Dashboard, log explorer, PDF report, email alerts, and SQLite run history.

Tools and Expected Output

The system uses Python, Streamlit, Pandas, Scikit-learn, SQLite, and ReportLab. The expected output is a working dashboard plus sample data, tests, and report-ready evidence.

Feasibility and Risk Analysis

Risk	Mitigation
Log format variation	Normalize common fields and preserve raw messages.
False positives	Use grouped alerts, confidence labels, and a labeled safe evaluation set.
Demo machine setup	Provide requirements.txt, run_sentinellite.ps1, and smoke tests.
Academic integrity	Cite open-source libraries and ensure the student can explain every design decision.